

README

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2026-10-08

Formative Assessment 7: Campus-Related Statistical Analysis

Part 1: Campus-Related Issues Using Exponential Distribution

Scenario

The selected campus-related problem is the waiting time between printing services at printing services located at the 2nd floor library. Rather than measuring the time between customers arriving, the observation focused on the time interval between one printing service being completed and the next printing service being completed.

The exponential distribution was selected because it can be used to model waiting times between events that occur randomly over time.

Data Collection

I, as an observer, quietly settled near the printing area from 14:09 to 15:08 (2:09pm - 3:08pm) on October 1. A clock app from my phone was used for stopwatch, recording the duration between completed printing services. By the end of observing, a total of 35 data were collected. The observations were recorded in seconds using Excel and were later analyzed using R.

The observations represent intervals between printing services rather than individual customer arrival times.

The mean waiting time was calculated from the 35 recorded intervals. The rate parameter, λ , was then estimated as the reciprocal of the mean waiting time.

The exponential probability density function (PDF) and cumulative distribution function (CDF) were also calculated using R. In addition, the probabilities of a printing service occurring within one, two, and five minutes were determined.

Results

The analysis produced the following results:

Number of observations: 35

Mean waiting time: 100.4829 seconds

Mean waiting time: 1.6747 minutes

Standard deviation: 103.3149 seconds

Lambda per second: 0.009952

Lambda per minute: 0.597117

Expected waiting time: 100.4829 seconds

Expected waiting time: 1.6747 minutes

The calculated probabilities were:

Within 1 minute: 44.96%

Within 2 minutes: 69.71%

Within 5 minutes: 94.95%

The coefficient of variation was 1.0282. This is close to 1, which is consistent with an important characteristic of the exponential distribution, where the theoretical coefficient of variation is 1.

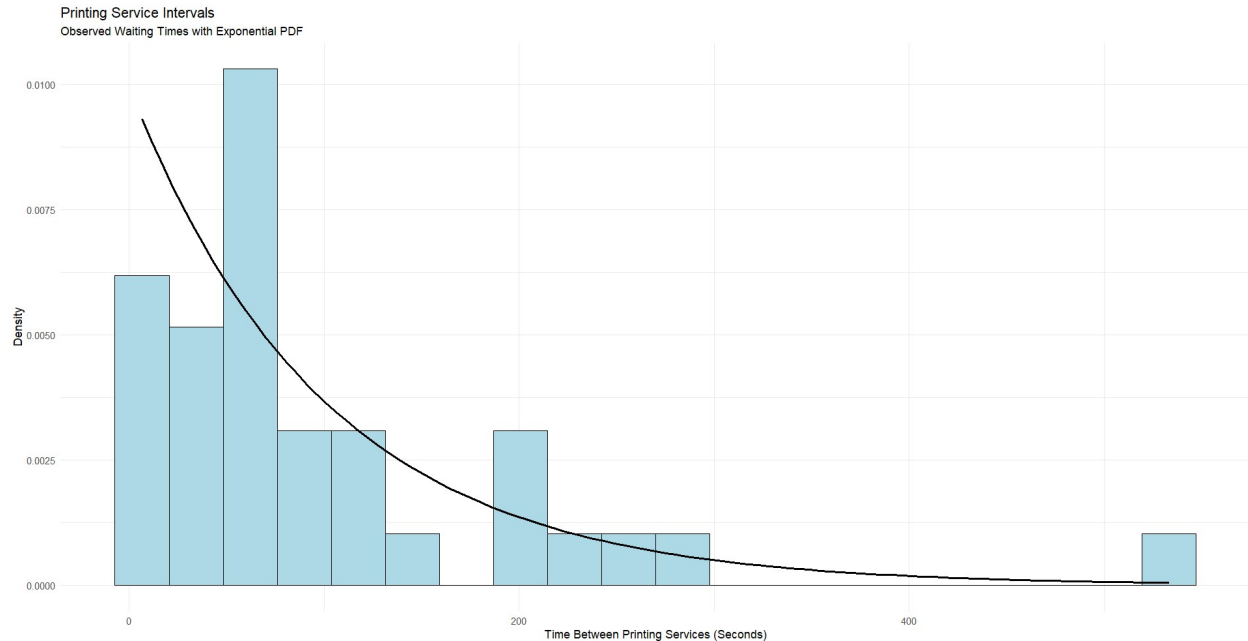


Figure 1: Histogram with Exponential PDF

Interpretation

The results indicate that the average interval between completed printing services was approximately 100.48 seconds, or 1.67 minutes. Based on the fitted exponential model, there is a 44.96% probability that another printing service will be completed within one minute. This probability increases to 69.71% within two minutes and 94.95% within five minutes.

These results suggest that although printing services generally occur within a relatively short interval, the actual waiting time can vary considerably. The standard deviation of 103.31 seconds is close to the mean waiting time, which is consistent with the variability expected from an exponential waiting-time model.

The histogram shows that most printing service intervals were short, while only a few took much longer. The data are right-skewed, which is a common shape for an exponential distribution. The exponential curve generally follows the pattern of the observed data, so the model provides a reasonable approximation of the waiting times.

The Q-Q plot shows that most of the observed values are reasonably close to the reference line, especially for the smaller and middle waiting times. However, the larger waiting times deviate from the line, with the longest interval being noticeably higher than the theoretical value. This means the exponential model is a reasonable approximation, but it does not perfectly fit the extreme observations.

In practical terms, the results can provide a general estimate of how frequently printing services are completed at the library. This may help students form reasonable expectations about the time between completed printing jobs. However, the results should be interpreted as an estimate based on the observed period and may not represent all operating periods of the library.

Part 2. Applying Normal Distribution on Campus

Scenario

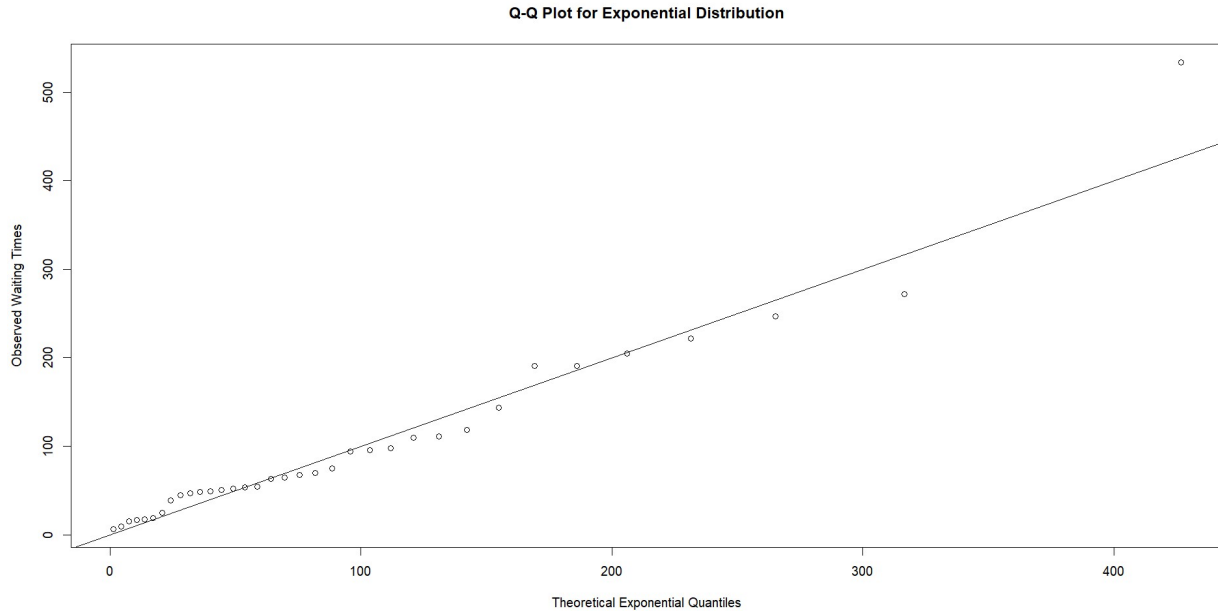


Figure 2: Q-Q Plot for Exponential Distribution

The selected variable is the amount of time people spent in Freedom Park. The purpose of this observation was to examine how long people typically remained in the area and determine whether the observed data followed a pattern that could reasonably be described using a normal distribution.

Data Collection

Observations were conducted in the Freedom Park, a popular spot among the students, on October 1 from 15:19 to 16:56 (3:19pm - 4:56pm). I keep track of the records using Samsung Notes on my phone. Data include the number of people observed, their starting time to stay in that area, and the time they left. These were input in Excel to compute for the total minutes people spent.

The observation was conducted without directly interacting with or disturbing the people. I quietly recorded the necessary information without intentionally focusing attention on individuals.

A total of 53 people were represented in the dataset. When multiple people had the same recorded duration, the frequency was retained in the analysis so that each person contributed to the statistical calculations.

The frequency of each recorded duration was organized into a frequency distribution table. The mean, standard deviation, and median were then calculated.

The mean time spent was used as μ , while the standard deviation was used as σ . The observed data were also compared with the theoretical normal distribution.

The percentage of observations within one, two, and three standard deviations of the mean was calculated. An interquartile range (IQR) method was also used to identify potential outliers.

Results

The analysis produced the following descriptive statistics:

Number of people: 53

Mean time spent: 16.8868 minutes

Standard deviation: 16.6001 minutes

Median: 11 minutes

Minimum: 1 minute

Maximum: 60 minutes

Frequency Distribution Table

```
part2 <- read.csv("D:/r fas/fa7/data2.csv")
part2 <- part2[1:38, 1:4]
totalp <- sum(part2$PEOPLE)

duration_data <- rep(
  part2$MINUTES_SPENT,
  part2$PEOPLE
)

frequency_table <- as.data.frame(table(duration_data))

names(frequency_table) <- c(
  "MINUTES_SPENT",
  "FREQUENCY"
)

frequency_table$MINUTES_SPENT <-
  as.numeric(as.character(frequency_table$MINUTES_SPENT))

frequency_table$PERCENTAGE <-
  frequency_table$FREQUENCY / totalp * 100

print(frequency_table)
```

##	MINUTES_SPENT	FREQUENCY	PERCENTAGE
## 1	1	6	11.320755
## 2	2	4	7.547170
## 3	3	1	1.886792
## 4	4	1	1.886792
## 5	5	2	3.773585
## 6	6	4	7.547170
## 7	8	1	1.886792
## 8	9	4	7.547170
## 9	10	2	3.773585
## 10	11	2	3.773585
## 11	12	6	11.320755
## 12	19	3	5.660377
## 13	21	3	5.660377
## 14	25	2	3.773585
## 15	26	3	5.660377
## 16	35	1	1.886792
## 17	36	1	1.886792
## 18	41	2	3.773585
## 19	46	1	1.886792
## 20	55	1	1.886792
## 21	60	3	5.660377

The percentage of observations within each standard-deviation range was:

Within 1sd: 83.02%

Within 2sd: 92.45%

Within 3sd: 100.00%

For comparison, the theoretical percentages for a perfectly normal distribution are approximately 68.27%, 95.45%, and 99.73% within one, two, and three standard deviations, respectively.

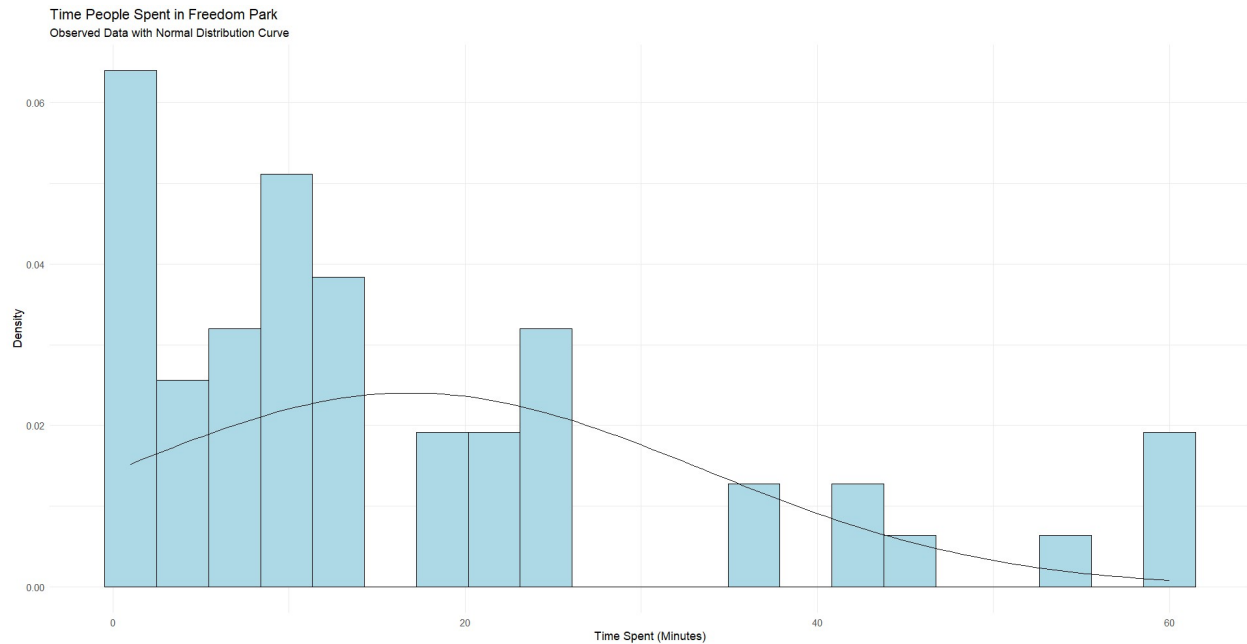


Figure 3: Histogram with Normal Distribution Curve

Distribution Shape and Symmetry

The distribution is not symmetric and is likely right-skewed. The mean of 16.89 minutes is greater than the median of 11 minutes. This difference suggests that the longer durations pull the mean toward the right side of the distribution.

The observed data also contain several relatively long stays compared with the more frequent shorter stays. Therefore, the data do not appear to follow a perfectly normal distribution.

Outliers

Using the $1.5 \times \text{IQR}$ rule, the first quartile (Q1) was 5 minutes and the third quartile (Q3) was 25 minutes. The IQR was therefore 20 minutes.

The lower and upper bounds were calculated as -25 minutes and 55 minutes, respectively. Three observations were identified as outliers, all having a duration of 60 minutes.

These observations indicate that a small number of people remained in the park substantially longer than most of the observed individuals.

Interpretation and Usefulness

The results show that people spent an average of approximately 16.89 minutes in the campus park, although the relatively large standard deviation of 16.60 minutes indicates considerable variation in the amount of time spent there.

The right-skewed shape suggests that shorter visits were more common, while a smaller number of people stayed for considerably longer periods. The three 60-minute observations contribute to the longer right tail of the distribution.

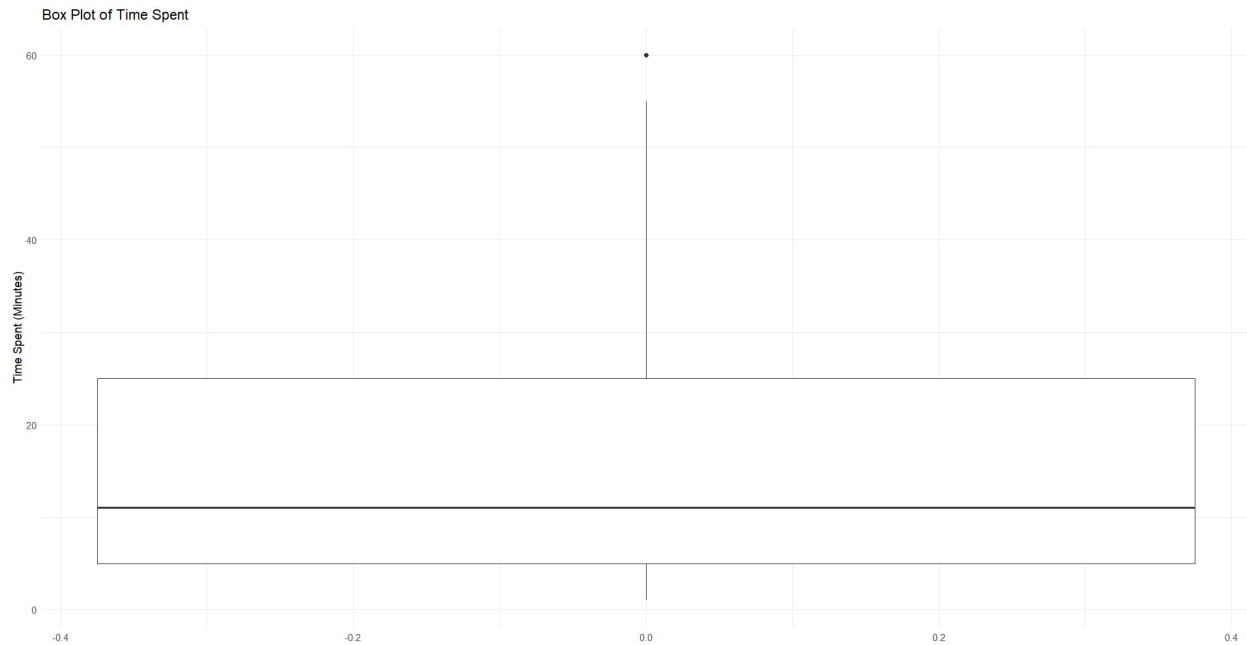


Figure 4: Box Plot

The histogram indicates that time spent in Freedom Park is not normally distributed. Most individuals spent relatively short periods in the park, while a smaller number stayed considerably longer. The distribution is positively (right) skewed, with values extending to approximately 60 minutes. The normal distribution curve does not closely match the observed data.

The box plot shows that the median time spent in Freedom Park was approximately 10–11 minutes. The middle 50% of observations were roughly between 5 and 25 minutes. The distribution is positively skewed, with a long upper tail. There is also an apparent high-value outlier at approximately 60 minutes, indicating that one observation involved a substantially longer stay than most visitors.

The data may be useful for understanding how students and other campus users utilize the park. For example, the information could help provide an estimate of typical visit duration and could contribute to decisions regarding seating, space allocation, or other improvements to the area. However, because the observations were collected during a particular observation period, the results should not be assumed to represent every person who uses the campus park at all times.

Summary and Conclusion

The two parts demonstrate how probability distributions can be applied to campus-related observations. The printing-service data produced an average interval of approximately 1.67 minutes between completed services, with the exponential model estimating a 94.95% probability of another service occurring within five minutes. The campus-park data showed an average stay of approximately 16.89 minutes and a right-skewed distribution rather than a perfectly symmetric normal distribution.

Overall, the analysis demonstrates how collected campus observations can be summarized and interpreted using probability and statistical methods.